Getting Started with L76K GNSS Module for SeeedStudio XIAO



Get One Now

Introduction

L76K GNSS Module for SeeedStudio XIAO is a Multi-GNSS (Global Navigation Satellite System) module compatible with all XIAO development boards, supports GPS, BeiDou (BDS), GLONASS and QZSS systems, allows multi-system combined or single-system independent positioning. It also supports AGNSS function, built-in low-noise amplifier and sound surface filter, and provides a good positioning experience of fast, accurate, high-performance.

The module comes with a high-performance active GNSS antenna intended to cover GPS L1 C/A, BeiDou B1 and GLONASS L1 bands. The design also has a tiny bright green LED for indicating the 1PPS output on fix.

Features

- **Enhanced Reception:** Built-in Low Noise Amplifier and Surface Acoustic Wave Filter for improved sensitivity and noise reduction
- **High Precision:** 32/72 channels, -162dBm tracking, -160dBm re-acquisition sensitivity
- **Energy Efficiency**: 41mA tracking/acquisition, 360μA standby
- **Multi-GNSS Systems**: Powered by Quectel L76K, supporting GPS, BeiDou, GLONASS and QZSS
- **Ceramic Antenna:** Enhanced signal reception, superior to traditional antennas.

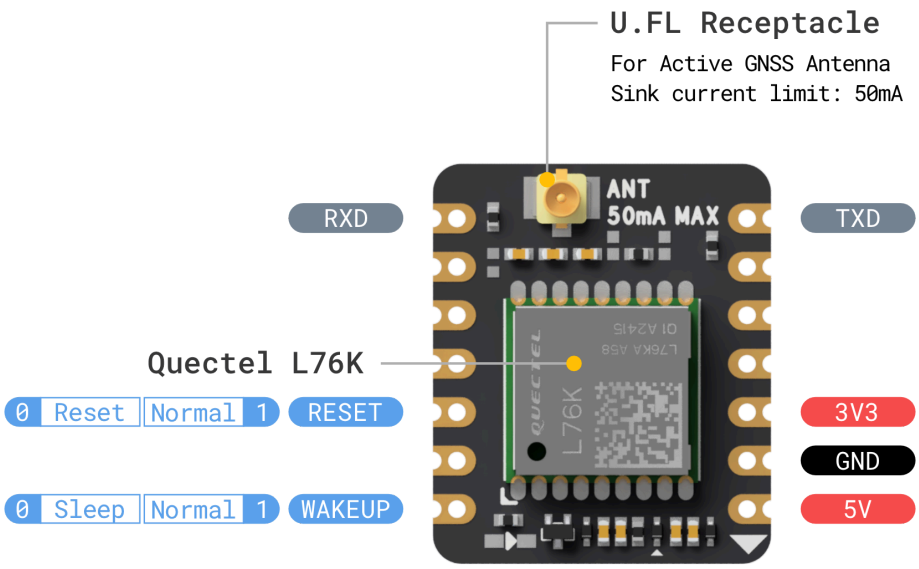
Specification

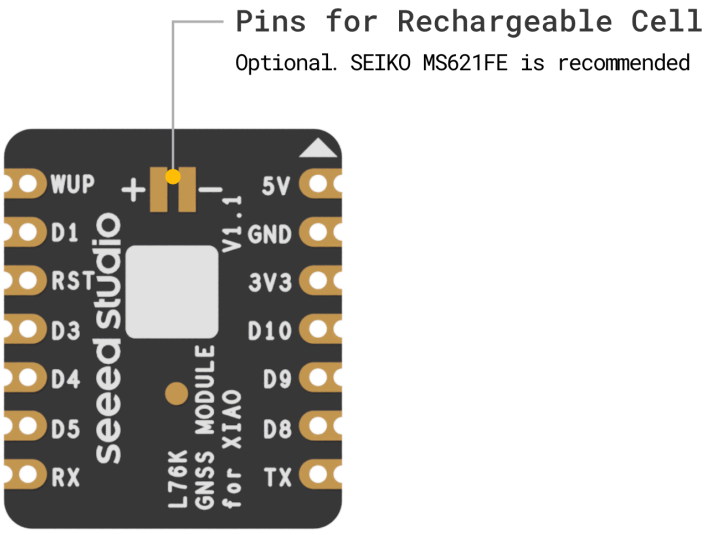
Item	Detail
GNSS bands	GPS L1 C/A: 1575.42MHz GLONASS L1: 1602MHz BeiDou B1: 1561.098MHz
Channels	32 tracking ch/72 acquisition ch
TTFF (Time To First Fix)	Cold Starts: 30s(w/o AGNSS), 5.5s(w/ GNSS) Hot Starts: 5.5s(w/o AGNSS), 2s(w/ AGNSS)

Sensitivity	Auto-aquisition: -148dBm Tracking: -162dBm Re-acquisition: -160dBm
Accuracy	Position: 2.0m CEP Velocity: 0.1m/s Acceleration: 0.1m/s^2 Timing: 30ns
UART Interface	Baud Rate: 9600~115200bps(9600bps default) Update Rate: 1Hz(default), 5Hz(Max.) Protocol: NMEA 0183, CASIC proprietary protocol
Antenna	Type: Active Antenna Operating Frequency: 1559–1606MHz Coaxial Cable: RF1.13 Length=10cm Cable Connector: U.FL Plug RA
Current consumption(w/ active antenna)	Auto-aquisition: 41mA Tracking: 41mA Standby: 360uA
Dimension	18mm x 21mm

Hardware Overview

Before we start, we can refer to the following pictures to understand the pin design of the L76K GNSS Module for SeeedStudio XIAO to facilitate our understanding of the function of the module.





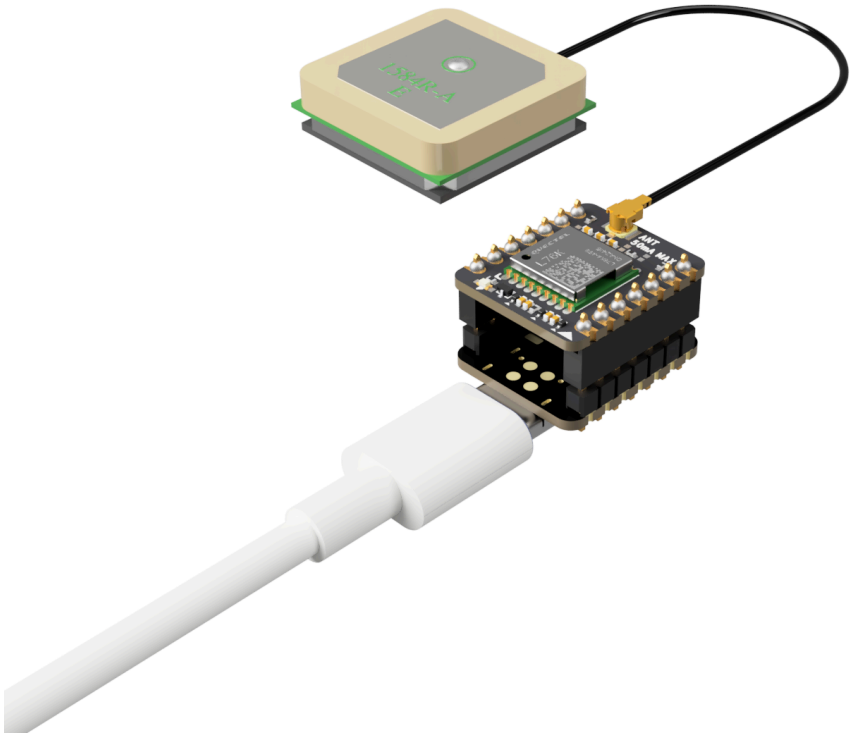
Getting Started

Hardware Preparation

To fully experience the capabilities of the L76K GNSS Module, we recommend pairing it with a motherboard from our XIAO series. *Any of the following XIAO models* would be compatible for use with the L76K GNSS Module.

Seeed Studio XIAO SAMD21	Seeed Studio XIAO RP2040	Seeed Studio XIAO nRF528 (Sense)
		
Get One Now	Get One Now	Get One Now

Before using this module on a XIAO motherboard, you need to install the header sockets on the module and plug the active GNSS antenna onto the module. When connecting to the XIAO, please pay special attention to the installation direction of the module, please do not plug it in backwards, otherwise it is likely to burn the module or the XIAO.



CAUTION

Please pay special attention to the installation direction of the module, please do not plug it in backwards, otherwise it is likely to burn the module or the XIAO.

Software Preparation

To use the L76K GNSS Module for SeeedStudio XIAO, we need to program the XIAO series. The recommended programming tool is the Arduino IDE, and you need to configure the Arduino environment for the XIAO and add the on-board package.

TIP

If this is your first time using Arduino, we highly recommend you to refer to Getting Started with Arduino.

Step 1. Download and Install the stable version of Arduino IDE according to your operating system

Download Arduino IDE

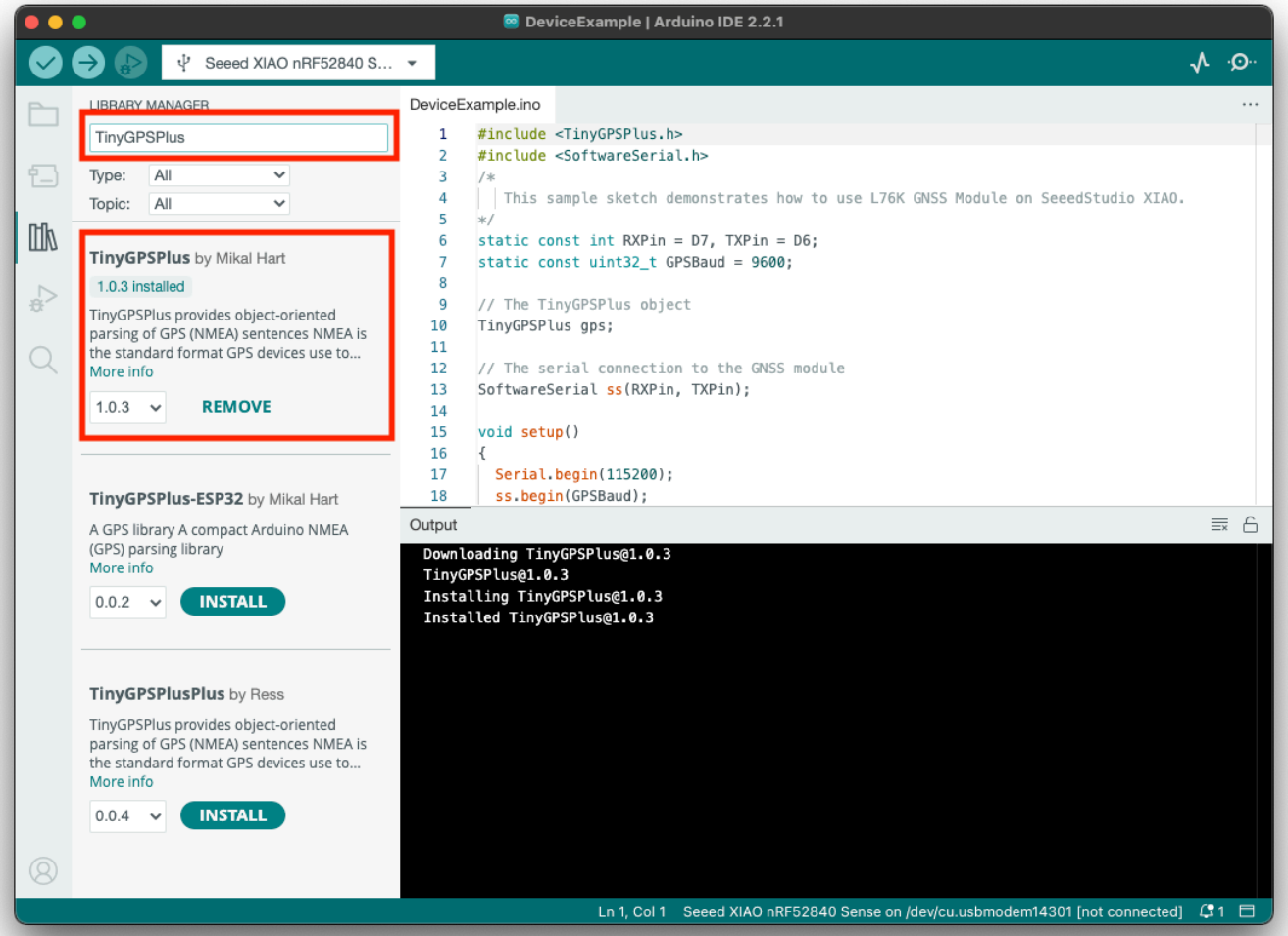
Step 2. Launch the Arduino application

Step 3. Configure the Arduino IDE for the XIAO you are using

- If you want to use **Seeed Studio XIAO SAMD21** for the later routines, please refer to **this tutorial** to finish adding.
- If you want to use **Seeed Studio XIAO RP2040** for the later routines, please refer to **this tutorial** to finish adding.
- If you want to use **Seeed Studio XIAO nRF52840** for the later routines, please refer to **this tutorial** to finish adding.
- If you want to use **Seeed Studio XIAO ESP32C3** for the later routines, please refer to **this tutorial** to finish adding.
- If you want to use **Seeed Studio XIAO ESP32S3** for the later routines, please refer to **this tutorial** to finish adding.

Step 4. Add the TinyGPSPlus library to Arduino

First, you need to search and download the latest version **TinyGPSPlus** library in the Arduino IDE.



Application Demo

Example 1: Reading and Displaying GNSS Data

Once the hardware and software are ready, we start uploading our first example program. The L76K GNSS Module prints the GNSS information via the serial port every 1 second after powering up. In this example, we will use **TinyGPSPlus** library to parse the NMEA sentences received from the module and print the results including atitude, longitude and time to the Arduino IDE's Serial Monitor.

Here is the source code:

```
#include <TinyGPSPlus.h>
#include <SoftwareSerial.h>
```

```
/*
   This sample sketch demonstrates how to use L76K GNSS Module on
SeeedStudio XIAO.
*/
static const int RXPin = D7, TXPin = D6;
static const uint32_t GPSBaud = 9600;

// The TinyGPSPlus object
TinyGPSPlus gps;

// The serial connection to the GNSS module
SoftwareSerial ss(RXPin, TXPin);

void setup() {
  Serial.begin(115200);
#ifdef ARDUINO_SEEED_XIAO_RP2040
  pinMode(D2,OUTPUT);
  digitalWrite(D2,1);
  pinMode(D0,OUTPUT);
  digitalWrite(D0,1);
#endif
  ss.begin(GPSBaud);

  Serial.println(F("DeviceExample.ino"));
  Serial.println(F("A simple demonstration of TinyGPSPlus with L76K GNSS
Module"));
  Serial.print(F("Testing TinyGPSPlus library v. "));
  Serial.println(TinyGPSPlus::libraryVersion());
  Serial.println(F("by Mikal Hart"));
  Serial.println();
}

void loop() {
  // This sketch displays information every time a new sentence is
correctly encoded.
  while (ss.available() > 0)
    if (gps.encode(ss.read()))
      displayInfo();

  if (millis() > 5000 && gps.charsProcessed() < 10) {
    Serial.println(F("No GPS detected: check wiring."));
    while (true);
  }
}
```

```
void displayInfo() {
  Serial.print(F("Location: "));
  if (gps.location.isValid()) {
    Serial.print(gps.location.lat(), 6);
    Serial.print(F(","));
    Serial.print(gps.location.lng(), 6);
  } else {
    Serial.print(F("INVALID"));
  }

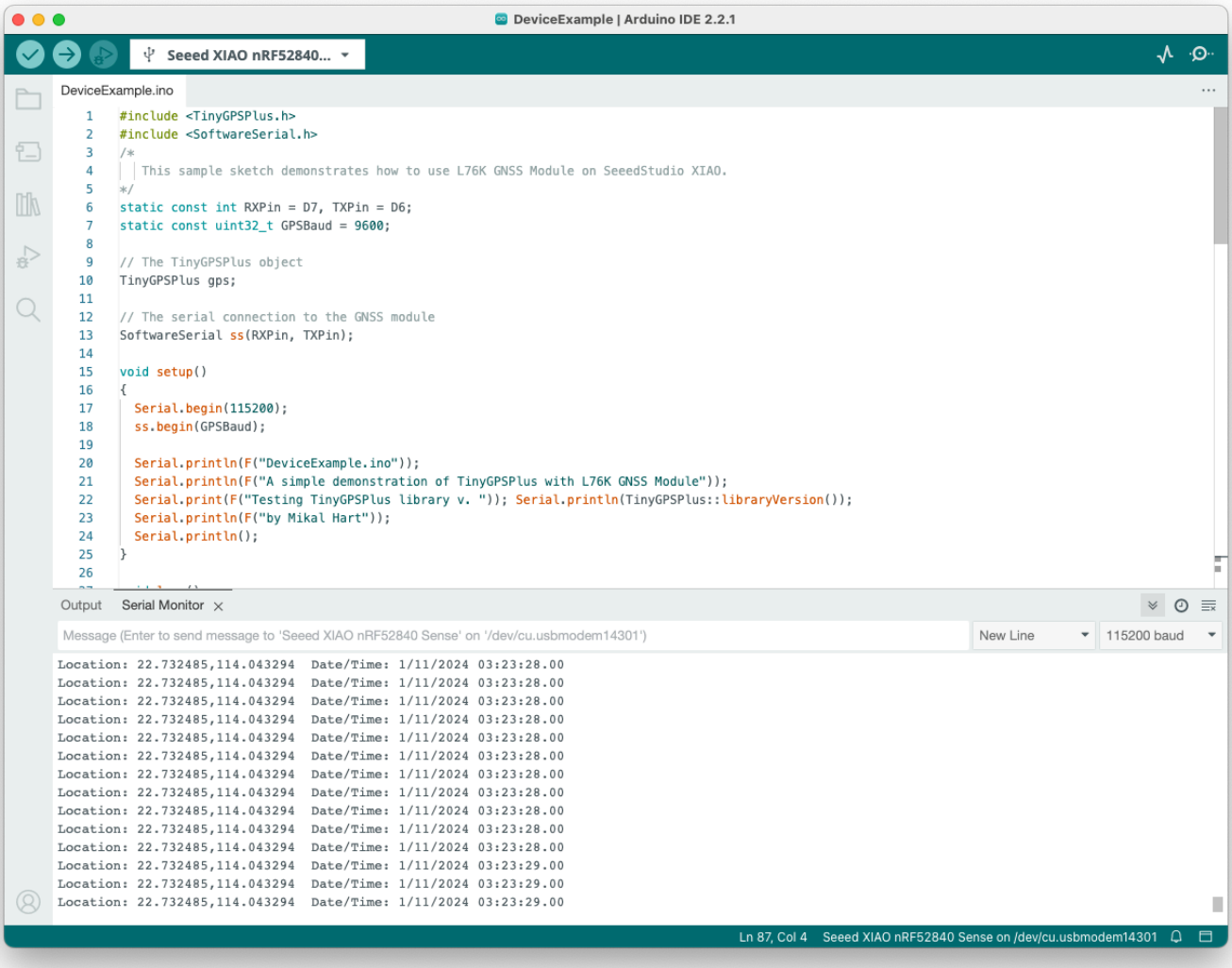
  Serial.print(F("  Date/Time: "));
  if (gps.date.isValid()) {
    Serial.print(gps.date.month());
    Serial.print(F("/"));
    Serial.print(gps.date.day());
    Serial.print(F("/"));
    Serial.print(gps.date.year());
  } else {
    Serial.print(F("INVALID"));
  }

  Serial.print(F(" "));
  if (gps.time.isValid()) {
    if (gps.time.hour() < 10) Serial.print(F("0"));
    Serial.print(gps.time.hour());
    Serial.print(F(":"));
    if (gps.time.minute() < 10) Serial.print(F("0"));
    Serial.print(gps.time.minute());
    Serial.print(F(":"));
    if (gps.time.second() < 10) Serial.print(F("0"));
    Serial.print(gps.time.second());
    Serial.print(F("."));
    if (gps.time.centisecond() < 10) Serial.print(F("0"));
    Serial.print(gps.time.centisecond());
  } else {
    Serial.print(F("INVALID"));
  }

  Serial.println();
}
```

Just select the XIAO you are using and the port number where the XIAO is located, compile and upload it.

Make sure that the L76K GNSS Module is placed outdoor where good GNSS signals can be received. Upload the code to your XIAO and wait a few minutes, you should see the information displayed on the serial monitor.



This code uses the TinyGPSPlus library to read data from the L76K GNSS module via a serial connection and displays valid location information and date/time on the serial monitor.

Resources

- **PDF**: L76K GNSS Module for Seeed Studio XIAO Schematic
- **PDF**: Quectel_L76K_GNSS_协议规范_V1.0
- **PDF**: Quectel_L76K_GNSS_Protocol_Specification_V1.1

- **GitHub**: Seeed_L76K-GNSS_for_XIAO

Troubleshooting

Details

Why doesn't the GNSS information display on the serial monitor?

Details

Tech Support & Product Discussion

Thank you for choosing our products! We are here to provide you with different support to ensure that your experience with our products is as smooth as possible. We offer several communication channels to cater to different preferences and needs.

*Last updated on **Oct 14, 2024** by **Spencer***

Loading comments…